Building Data Apps In Python With Streamlit

Session 3: Graphics

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Exercise: What Graphics?

At the start of the first session I asked:

- *What do you want to build after completing this course?*

Now answer this question:

- *What graphs do you want your app to have?*

Course Agenda
(Big Picture)

1. Session 1: Setup & Basics
2. Session 2: Inputs
3. Session 3: Graphics
4. Session 4: UI
5. Session 5: Deployment

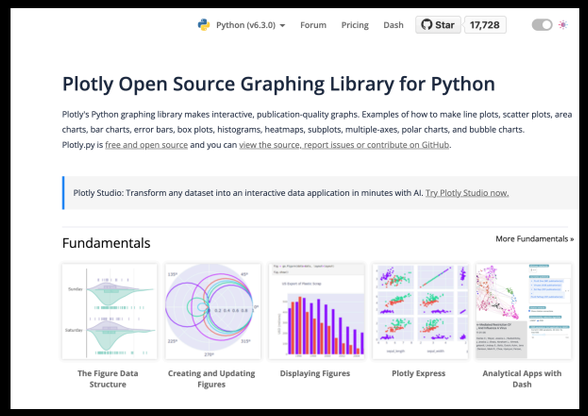
We now have an app that lets users select a state, and we show them a filtered dataframe. In this session we'll add graphics to our app.

Please Open

- 3-graphics/graphs.ipynb
- 3-graphics/graph_app.py

Plotly

- Popular library for interactive graphics
- "Plotly Express"
  - import plotly.express as px
  - px.line
  - px.choropleth
- Downloaded 31M times last month



Plotly is a popular Python library for making interactive graphics. This lets users hover their mouse over a point and see more information about it. It's common to use interactive graphics when making web apps.

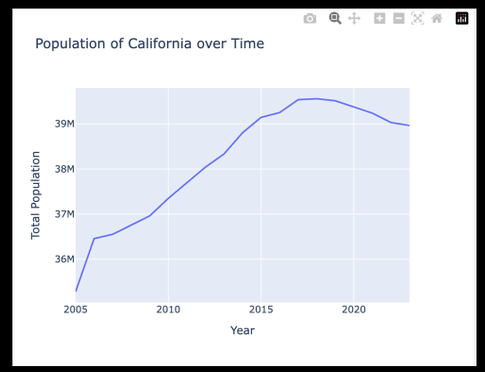
Plotly has a module called "Plotly Express" that lets you easily make common graphics. Each function in that module has a descriptive name. We'll be using Plotly Express in this course.

px.line()

Graph Population over Time

	State	Year	Total Population	Median Household Income
72	California	2005	35278768	53629
73	California	2006	36457549	56645
74	California	2007	36553215	59948
75	California	2008	36756666	61021
76	California	2009	36961664	58931
77	California	2010	37349363	57708
78	California	2011	37691912	57287

```python
import plotly.express as px

px.line(
    df_ca,
    x="Year",
    y="Total Population",
    title="Population of California over Time"
)
```



How would we graph population over time? Let's start by reviewing the structure of our data.

We have a dataframe with columns like Year and Total Population. Plotly uses those column names when making the x- and y-axis.

We start by importing plotly express as px. Then we call px.line. The first argument is our dataframe. Then we specify the column we want to be the x-axis. Then we specify the column we want to be the y-axis. Then we specify the title.

Plotly Graphs in Streamlit

st.plotly_chart

Version 1.49.0

Display an interactive Plotly chart.

Plotly is a charting library for Python. The arguments to this function closely follow the ones for Plotly's `plot()` function.

To show Plotly charts in Streamlit, call `st.plotly_chart` wherever you would call Plotly's `py.plot` or `py.iplot`.

Calling px.line() in a Streamlit app won't render the plot. You need to call st.plotly_chart.

Plotly Graphs in Streamlit

```
fig = px.line(...)
st.plotly_chart(fig)
```

This is the most common pattern for rendering plotly charts in Streamlit. You call a plotly express function like normal. But you store the result in a variable call fig. And then you pass fig to st.plotly_chart. This is what I'll ask you to do in the next exercise.

f-Strings

f-Strings

For the next exercise, it will help to know about "f-strings".

```
state = "New York"
title = f"Graph for {state}"
title
```

✓ 0.0s

```
'Graph for New York'
```

In the next exercise, I'm going to ask you to make a graph that updates based on the state that the user selects. You'll probably want the title of the graph to update with the name of the state as well. To do that, it's helpful to know about f-strings.

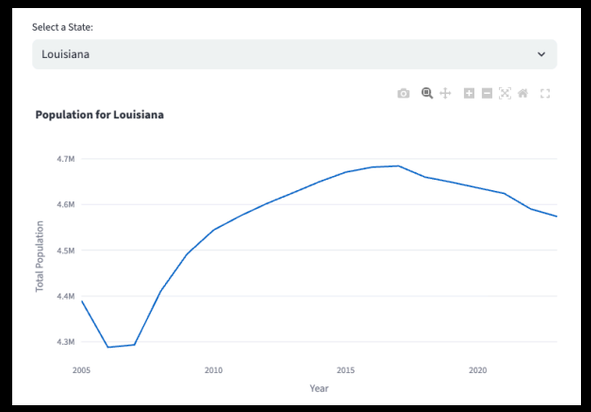
The "f" stands for "formatted". To make an f-string, put the letter 'f' in front of the quotes. Then put a variable inside curly braces. The value inside the variable will become part of the string.

Exercise: Plotly Graphs in Streamlit

Update `graph_app.py` to create a line graph of the population of the selected State.

Solution

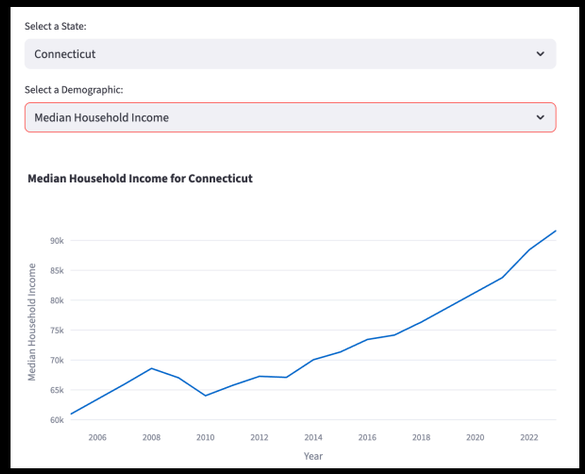
```
fig = px.line(df_state,
    x="Year",
    y="Total Population",
    title=f"Population for {state}")
st.plotly_chart(fig)
```



And here's the result.

Exercise: Select & Graph a Demographic

- Add a new select box to graph_app.py.
  - Options: Total Population and Median Household Income.
- Store the value returned from that selectbox in a variable called demographic.
- Use the value of demographic as the value for y in your line graph.



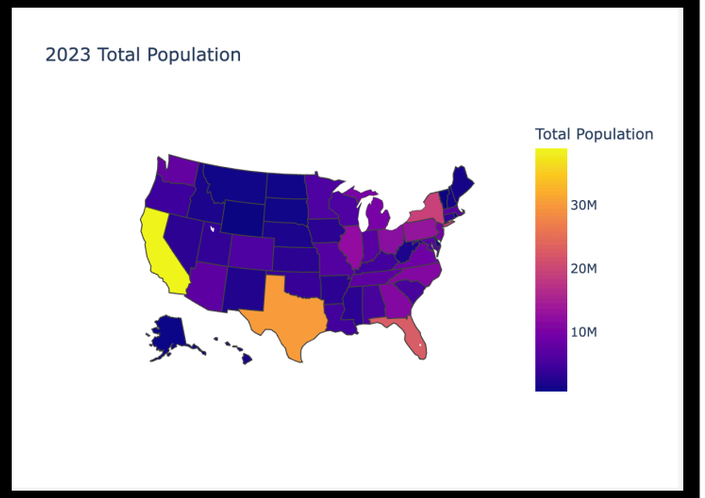
In some sense this is the final exam, because if you can do this, you can do anything with Streamlit.

Solution

```python
import streamlit as st
import pandas as pd
import plotly.express as px

st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

state = st.selectbox("Select a State:", df["State"].unique())
demographic = st.selectbox(
    "Select a Demographic:", ["Total Population", "Median Household Income"]
)

mask = df["State"] == state
df_state = df[mask]

fig = px.line(df_state, x="Year", y=demographic, title=f"{demographic} for {state}")
st.plotly_chart(fig)

st.dataframe(df_state)
```

Choropleth Maps

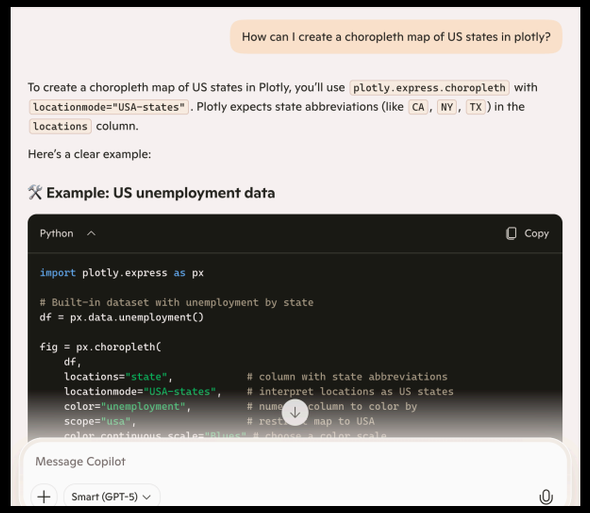
- Shows locations such as states
- Uses color to express values
- Different than a heat map



Our data lends itself to choropleth maps. Choropleths show areas (like states) and express values for those areas with color.

Exercise: Copilot choropleth code

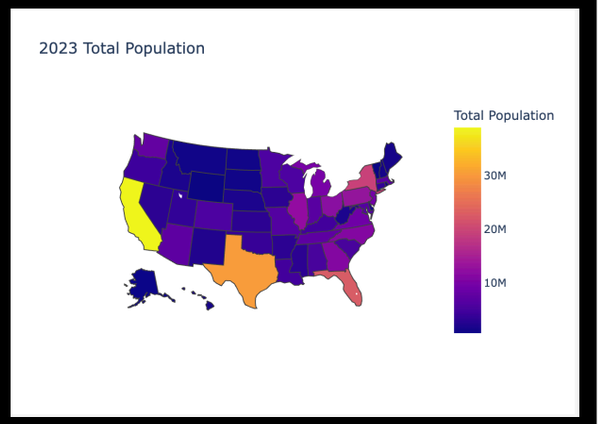
- Prompt: "*How can I create a choropleth map of US states in plotly?*"
- Take a minute to read the answer.



Like with select boxes in the last lesson, copilot is great for getting starter code. When you broadly know what you want, it's great for helping to fill in the blanks.

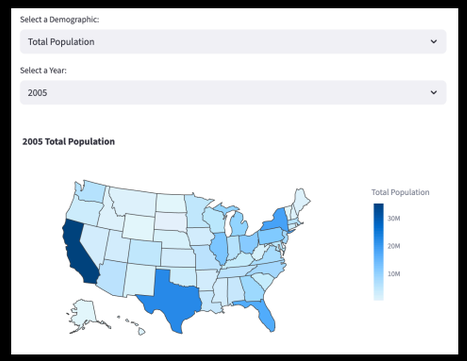
Example Choropleth Code

```python
px.choropleth(
    df_2023,
    locations="State Abbrev",  # Column for region
    locationmode="USA-states",
    color="Total Population",  # Column for color
    scope="usa",
    title="2023 Total Population",
    color_continuous_scale="viridis",
)
```



The notebook has this code, which shows you how to write code that filters a dataframe to a specific variable and year, and create a map of it. And it also has an exercise for creating a dropdown in the app that lets users select the demographic that they want to see.

Exercise: Choropleth Map



1. Copy the above code, verbatim, to the app. Verify that it works.
2. Connect the map to the `demographic` selectbox. Do the results surprise you?
3. Create a new selectbox that lets the user select which `Year` of data to map.
4. Connect the year selectbox to the map.

The second question of your final exam.

At the start, the map will not be connected to any UI widgets. Gradually connect it: first demographic then year.

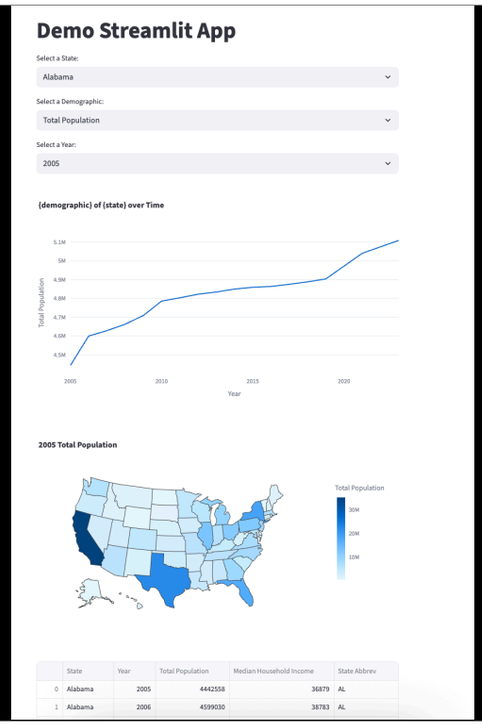
Solution

```python
demographic = st.selectbox(
    "Select a Demographic:", ["Total Population", "Median Household Income"]
)
year = st.selectbox("Select a Year:", df["Year"].unique())

mask = df["Year"] == year
df_choro = df[mask]

fig = px.choropleth(
    df_choro,
    locations="State Abbrev",  # Column for region
    locationmode="USA-states",
    color=demographic,  # Column for color
    scope="usa",
    title=f"{year} {demographic}",
)
st.plotly_chart(fig)
```

The year selectbox works just like the state selectbox - use .unique() because 2020 is missing.

Up Next: Layouts

- Organize the User Interface (UI) a bit better
- Move select boxes into columns
- Move graphs into tabs



The app is interesting but a bit too crowded. In the next section we'll learn how to organize the graphs and selectboxes a bit better.

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Reiterate course outline, and that this section is the most important skill when working with Streamlit.